



(Media Attention to) Misinformation Can Undermine Trust in Scientists

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Abstract

Could news coverage of misinformation be harmful? Across two studies on U.S. citizens, we examine whether news coverage of misinformation generates misperceptions and decreases levels of trust in information institutions (i.e., media, professors, and scientists) and whether its effects can be comparable to those of exposure to untrustworthy content. We rely on an online experiment using mock social media posts (Study 1, N=1,670) and also on online behavioral tracking data paired with over-time survey self-reports (Study 2, N=804). Study 1 finds that exposure to both actual misinformation *and* the coverage of misinformation affect misperceptions, but does not decrease trust. Study 2 presents evidence that behaviorally tracked visits to untrustworthy sites and exposure to news coverage of misinformation—although relatively rare—do not affect misperceptions, but both predict lower levels of trust in scientists, with less consistent effects for media and university professors. These results support concerns that not only misinformation but also its coverage contribute to epistemic uncertainty by eroding confidence in credible sources of knowledge, and warrant further inquiry into the potential harms of news media's attention to misinformation.

Keywords Misinformation · Media coverage · Trust · Misperceptions · Survey experiments · Computational methods

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Introduction

What types of information lead the public to believe that vaccines are dangerous or that climate change is a hoax? Political scientists examining the origins of misperceptions often point to untrustworthy websites—sites that provide content that is false or (purposefully) misleading (Guess et al. 2020b)—as the origin (Aird et al. 2018; Clayton et al. 2020; Flynn et al. 2017). While untrustworthy sources are certainly problematic, only few people visit them (Aslett et al. 2022) and similarly small percentages encounter misinformation on social media (Grinberg et al. 2019; Guess et al. 2019). This raises the fundamental question of how it is possible that parts of the American public are misinformed (Jackson 2018) while—at the same time—so infrequently exposed to misinformation and untrustworthy sources.

We argue that, in addition to untrustworthy websites, it is important to consider other potential sources of misperceptions. Here, we call attention to one such plausible source, which has only recently received scholarly attention, namely the coverage of misinformation in the news media (Altay et al. 2024; Hameleers 2023; Hoes et al. 2024; Tandoc Jr et al. 2021; Thorson 2024; van der Meer et al. 2023; Van Duyn and Collier 2019). We distinguish between exposure to misinformation itself and exposure to news coverage of misinformation to advance the theoretical argument that news attention to misinformation, either reporting identified false claims made by politicians or presenting false claims accompanied by fact-checks, may have negative unintended consequences (Thorson 2024). We base this expectation on the well-known “familiarity effect” (Brashier and Marsh 2020; Swire et al. 2017; Zajonc 1968), according to which people are more likely to believe information they have encountered before, whether true or false.

Next to a well-informed public, citizens’ trust in information institutions is another important ingredient for the functioning of democracies. Here, we understand information institutions as organizations that play a central role in gathering, sharing, and producing information within a society (Tandoc Jr et al. 2021). Therefore, we consider trust in traditional and social media, journalists, scientists, fact-checkers, and academics as important outcomes. Exposure to misinformation should lower trust because such content frequently undermines the credibility of media and journalism (Eldridge II 2019; Rae 2021), sows uncertainty and confusion (Budak et al. 2025; Elswah and Howard 2020; Paul and Matthews 2016), and—by amplifying falsehoods—makes it harder to discern what and whom to trust (Altay and Acerbi 2024). Although often operationalized through exposure to untrustworthy sources, the underlying mechanism is tied to the nature and effects of the misleading or false content itself.

In theorizing the effects of news coverage of misinformation on trust, we extend research on priming and media trust (Van Duyn and Collier 2019) to argue that news discourse *about* misinformation may generate doubts about the validity of all information (Hoes et al. 2024; Tandoc Jr et al. 2021) and skepticism in generally trusted institutions (Bennett and Livingston 2018), as detailed below.

This project therefore examines an often-overlooked driver of misperceptions and institutional trust: not just misinformation itself, but news media coverage of misinformation.

To our knowledge, we are the first to examine the effects of *both* untrustworthy content and its journalistic coverage side-by-side across two studies—one experimental and one based on behavioral trace data—allowing us to probe how these different types of exposure shape public (mis)perceptions and trust in information institutions. Whereas most work has focused on misinformation alone or has relied on one method, we examine how exposure to falsehoods and to news coverage of misinformation—defined as news reporting that flags and discusses misinformation (Thorson 2024)—affect misperceptions and trust. Thus, we focus specifically on the type of content (misinformation versus its coverage), and not the sources that produce it.¹

In Study 1, we report a pre-registered survey experiment (see also Section D in the SI). A representative sample of 1,670 U.S. citizens was randomly assigned to read either news coverage of misinformation or actual misinformation. In Study 2, we combine individual-level web browsing data (> 11 million visits) with two survey waves ($n=804$ in the second wave), and also with publicly available lists of web domains and state-of-the-art machine learning methods that automatically classify whether news articles covered misinformation or not. We use these behavioral measures of actual exposure to untrustworthy sites and news coverage of misinformation to predict people's levels of misperceptions as well as trust in information institutions.

Our results offer critical insights. Study 1 finds that exposure to actual misinformation increases misperceptions while exposure to news coverage of misinformation decreases them, and that neither significantly affects trust. Study 2, however, shows that behaviorally tracked exposure to untrustworthy websites and to news coverage of misinformation does not predict increased misperceptions over time. Yet, even rare exposure to news coverage of misinformation and to untrustworthy sites decreases trust in scientists. We do not observe consistent associations for trust in journalists or university professors, suggesting that the impact of such exposures may be institution-specific, as we discuss later.

Although our findings vary by method and outcome, we offer a comprehensive assessment of the potential harms tied to both misinformation and the well-intentioned efforts of journalists and news media to address it. This paper therefore makes two key contributions. First, we move beyond the focus on either misinformation or its coverage in isolation and compare their effects side by side. Second, we leverage both experimental and observational behavioral data to examine whether these exposures affect misperceptions and trust in information institutions. By combining these two approaches, we speak to both internal and external validity and highlight how effects vary depending on method and context. Although our two studies do not yield consistent results for all outcomes—especially with regard to misperceptions—they suggest that even rare encounters with either misinformation or its coverage can lower trust in scientists. These findings complicate overly general claims about misinformation's harms and emphasize the need for more comparative, multi-method research in this area.

¹ As detailed below, we use exposure to untrustworthy websites as a proxy for exposure to misinformation in Study 2. Although it may not be a perfect proxy, as acknowledged in the discussion, it has been used extensively by researchers interested in misinformation (Grinberg et al. 2019; Guess et al. 2019, 2020b).

Theoretical Framework

Many scholars and public observers see political misperceptions as one of the key problems in contemporary democracies. This seems to be a particular issue in the U.S., which ranks second among thirteen countries on the Ipsos' misperceptions index (Jackson 2018) with, for example, few Americans believing vaccines to be effective. Misperceptions have concrete consequences, such as the growing “anti-vaxxer” movement that leads to outbreaks of preventable diseases and hampers immunity against COVID-19 (Johnson et al. 2020). Previous work examining the roots of misperceptions has mostly focused on untrustworthy sites. Although such sites are certainly problematic, our key contention in this project is that mere *attention to misinformation* in the news media can also have undesirable downstream effects on misperceptions and trust (Jungheer and Rauchfleisch 2022; Tandoc Jr et al. 2021; Thorson 2024; Van Duyn and Collier 2019). We first briefly discuss the effects of misinformation and untrustworthy websites and then outline the potential negative effects of news coverage of misinformation.

Misinformation, Untrustworthy Sites and Misperceptions

Previous work examining the roots of misperceptions has focused on individual pieces of fact-checked false information and/or on untrustworthy sites more broadly. Such sites—e.g., *The Daily Wire*, *Breitbart*, or *Infowars*—have been identified as key sources of misinformation, often publishing content that is false, (purposefully) misleading, unverified, or otherwise not supported by evidence (Allcott et al. 2019; Guess et al. 2020b).

Although some survey studies from countries other than the U.S. challenge the idea that self-reported exposure to misinformation or untrustworthy websites influences misperceptions (Altay et al. 2023b; Valenzuela et al. 2022) or find only a limited relationship between misinformation exposure and discernment (Altay et al. 2025b), most research suggests that exposure to misinformation and untrustworthy sites increases misperceptions by increasing people's familiarity with misinformation and leading them to internalize falsehoods (DiFonzo et al. 2016; Eveland and Cooper 2013; Pillai et al. 2023; Swire et al. 2017), the so-called familiarity effect, which we discuss in more detail in the following section.

This evidence comes primarily from studies that rely on self-reports or experimentally manipulated exposure (Murray et al. 2020; Pennycook and Rand 2019; Silverman and Singer-Vine 2016), designs that have known limitations related to recall or social desirability biases, and to ecological validity of forced exposure to an isolated piece of content, which users may never encounter in naturalistic settings. Addressing these problems, Guess et al. (2020a) use online traces and survey self-reports from a representative U.S. sample to show that exposure to untrustworthy websites predicts politically congenial misperceptions among Republicans, likely because such websites often have a conservative orientation (Benkler et al. 2018; Rae 2021). In a similar design, however, Weeks et al. (2021) match data from Facebook users'

browser histories with a survey measuring the accuracy of political beliefs, finding limited evidence that actual exposure to untrustworthy websites predicts false beliefs.

Because the mixed evidence may be due to different methods and samples, we present both experimental and online behavioral trace data. In Study 1, we examine these effects directly by exposing participants to fabricated false claims.² In Study 2, we approximate exposure to misinformation by measuring visits to untrustworthy websites. Although these sites may occasionally convey accurate information, their incentive structures and reputational histories make them far more likely to publish misleading or false content than mainstream news outlets (Benkler et al. 2018; Grinberg et al. 2019). We thus use visits to such domains as a behavioral proxy for likely misinformation exposure, rather than theorizing about the source characteristics themselves. The aforementioned theory on familiarity effects lead to the following hypothesis³:

Hypothesis 1: *Exposure to misinformation increases misperceptions.*

The Coverage of Misinformation and Misperceptions

Our key theoretical argument is that the news media may also be partly to blame for misinformed beliefs among the population. News media, whether partisan or not, increasingly focus on fake news and misinformation: Fig. 1 illustrates trends in news coverage of misinformation by plotting the occurrence of the terms “fake news,” “misinformation,” and “disinformation” in major U.S. newspapers over time (as archived by LexisNexis). This figure also shows the frequency of people *searching* for these terms on Google Search and Google News in the U.S. We see a remarkable increase in the popularity of these terms starting around the 2016 U.S. presidential election. For example, whereas there were 1,979 articles mentioning one of the terms in 2010, there were 9,012 such articles in 2017. This development is relevant because it clearly suggests that whereas most people do not see a lot of untrustworthy content, as past research finds, they might be regularly exposed to the *coverage of misinformation* in (mainstream) news media.

Following Thorson (2024), we define news coverage of misinformation as journalistic content that explicitly addresses the phenomenon of misinformation, including its spread, its dangers, and efforts to combat it. Although such coverage is typically corrective and well-intentioned, it still repeats or paraphrases false claims. There are

²We emphasize that our interest lies in the effects of exposure to misinformation regardless of its source, as noted below.

³Note that the formulation (but not the direction) of our hypotheses deviates from the pre-registered hypotheses. This is because for the pre-registration we administered hypotheses that spoke only to the survey experiment. We did not pre-register any hypotheses for the observational study. Hence, we formulated hypotheses that would be applicable to both studies and methods while keeping the directions of the hypotheses as pre-registered. For instance, in the pre-registration we hypothesized that the levels of misperceptions will be highest among participants exposed to actual misinformation, followed by those exposed to the coverage of misinformation, compared to participants in the control group. In this study, we translated this hypothesis from comparing groups to the effect of exposure to each type of information.

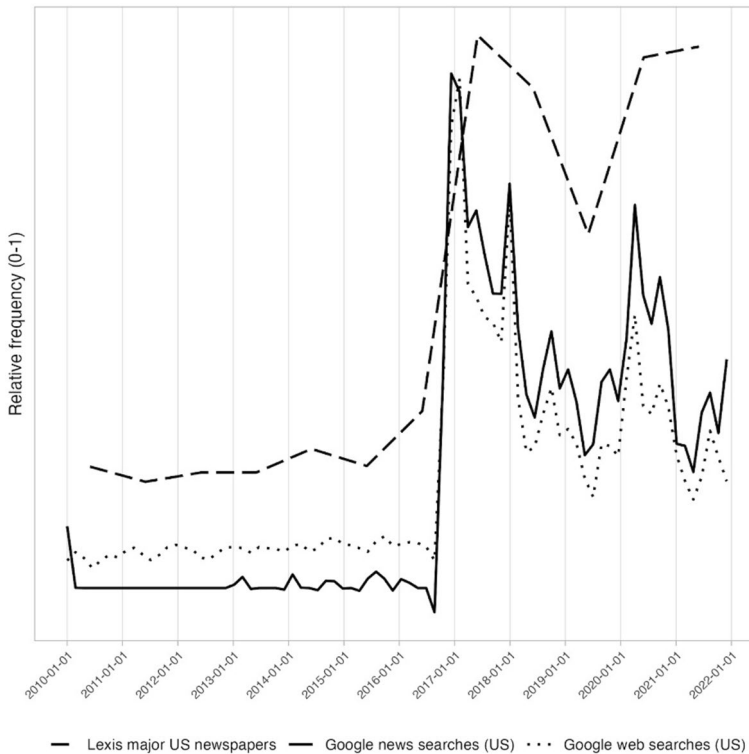


Fig. 1 Trends in the coverage of misinformation (2010–2021). Occurrence of terms “fake news,” “misinformation,” and “disinformation” in (1) newspaper articles archived by Lexis+ (<https://advance.lexis.com>), filtered to “Major U.S. Newspapers”, aggregated at the year level; (2) Google Search queries, aggregated at the month level; (3) Google News queries, aggregated at the month level. For comparability, frequencies were scaled to between 0 and 1

several theoretical reasons why news coverage of fake news, the perils of misinformation, or fact-checking may fuel misperceptions. Just like actual misinformation, news coverage thereof may foster misperceptions through the “familiarity” or “illusory truth” effect (Swire et al. 2017; Zajonc 1968), which “occurs when increased familiarity—often through repetition—gives rise to the illusion that information, even when implausible (Dechêne et al. 2010), is valid and inadvertently increases an individual’s belief” (Swire et al. 2017, p. 1948). That is, familiarity increases the ease of processing and retrieving information and serves as a heuristic or mental shortcut (Pennycook and Rand 2021), also referred to as fluency. Fluency, in turn, helps people make judgements more quickly, which people usually prefer over strategies that involve more cognitive energy (Brashier and Marsh 2020; Scholl et al. 2014; Unkelbach 2007; Unkelbach et al. 2019).

This phenomenon is well-documented (Dechêne et al. 2010; Hasher et al. 1977; Pennycook et al. 2018; Smelter and Calvillo 2020), also by meta-analyses (Dechêne et al. 2010). For instance, Swire et al. (2017) found that repetition of false claims increases the familiarity of “myths”, and these familiar myths are more likely to be deemed true. This happens even when the false claim is presented alongside a retrac-

tion, suggesting “that people tend to remember information while forgetting the context within which they encountered it” (Lazer et al. 2018, p. 1095), which can generate lingering negative effects on related attitudes or judgment, a phenomenon known as belief echoes (Thorson 2016). Similarly, Pennycook et al. (2018) found that even one single exposure to a piece of misinformation increases its perceived accuracy, even when misinformation is labeled as disputed by fact-checkers. Other studies also show that labeling content, such as politicians’ statements, as ‘true,’ ‘false,’ or ‘disputed’ may be insufficient to correct misperceptions or even be counterproductive (Wittenberg et al. 2020). We acknowledge that this evidence is not always consistent with studies on fact-checking specifically. On the one hand, meta-analyses show that fact-checking carries positive influence relative to control conditions (Chan et al. 2017; Walter et al. 2020). On the other hand, however, fact-checks often fail to reach their intended audience (Guess et al. 2020b) and—most importantly—provide incomplete protection against familiarity effects (Pennycook and Rand 2021; Pennycook et al. 2018; Thorson 2016). These apparent inconsistencies between the different studies are likely due to their slightly different foci and different methodologies (Brashier and Schacter 2020; Clayton et al. 2020; Freeze et al. 2021).

In addition and importantly, *news coverage of misinformation* is distinct from fact-checking labels studied to date. Our in-depth qualitative content analysis of misinformation coverage (see SI Section C.1) suggests that news media either repeat falsehoods while also mentioning they are false, or report falsehoods in the context of fact-checking, but with retractions not being prominently visible in longer media coverage.⁴ Similarly, based on a content analysis of U.S. media, Thorson (2024) shows that news coverage of misinformation often repeats false claims, sometimes in detail, even when those claims are ultimately debunked. Such repetition can inadvertently increase the familiarity and cognitive fluency of misinformation, making it more likely to be misremembered as true. Thus, Thorson’s work (2024) and our own qualitative findings support the concern that even well-intended news coverage of misinformation may increase misperceptions through repetition and familiarity. In sum, based on these frameworks and evidence, we pre-registered the following expectation:

Hypothesis 2: *Exposure to news coverage of misinformation increases in misperception.*

Consequences for Trust

In addition to a well-informed public, another ingredient needed in effective democracies is citizens’ trust in information institutions to fulfill their role (Kornberg 2005). For instance, if citizens distrust the news media, journalists will struggle to keep the public informed, which may lead to misinformed voting. We argue that both untrust-

⁴ For examples, see <https://www.vox.com/2018/4/18/17252410/jordan-peelee-obama-deepfake-buzzfeedor> <https://www.foxnews.com/politics/trump-claims-maxine-waters-iq-in-mid-60s-slams-fake-pocahontas-elizabeth-warren-in-rally-to-unseat-jon-tester>

worthy sites and news media coverage of misinformation can lower public trust in information institutions. Indeed, recent studies show that trust is declining and—germane to our argument—can be affected by (false) information to which people are exposed (Altay and Acerbi 2023; Van Duyn and Collier 2019).

Misinformation and untrustworthy sites may erode citizens' trust in the media and journalists because they challenge or antagonize traditional media to undermine journalism (Eldridge II 2019), endorse populist narratives that journalists are biased and incompetent and traditional outlets serve corrupt elites (Rae 2021), and, in general, aim to sow uncertainty, confusion, and distrust (Budak et al. 2025; Elswah and Howard 2020; Paul and Matthews 2016). Similarly, untrustworthy sites' cynical coverage of democratic institutions can cause people to distrust scientific organizations (Ognyanova et al. 2020). Untrustworthy sites and misinformation, then, may be an important source that delegitimizes the media and journalists as well as scientists and academics (Hopmann et al. 2015; Kleinnijenhuis et al. 2006; Nelson and Taneja 2018; Newman et al. 2016; Pfänder and Altay 2025). Indeed, one study finds that exposure to untrustworthy sites lowers trust in mass media among liberals, but increases trust for conservatives (Ognyanova et al. 2020), perhaps because the majority of untrustworthy sites produce right-leaning content (Guess et al. 2019). Another study shows that exposure to higher proportion of misinformation fuels distrust and the rejection of legitimate news (Altay et al. 2025b). That is, signs that news media might be untruthful may lead citizens to be skeptical and increase doubts in the media environment (Mayo 2019).

In addition, we posit that news coverage of misinformation may also lower trust. We offer both intuitive and theoretical reasons for this expectation.

Intuitively, when people are repeatedly told that misinformation is widespread and hard to detect, they may start to doubt not just unverified content but also trustworthy sources. Media coverage of misinformation often speaks directly to the difficulty of knowing whom and what to trust (Ternovski et al. 2022; Thorson 2024), and frequently features cues that signal attempted deception, such as phrases like “fake news is spreading” or “people are being misled”—which can heighten general skepticism (Altay et al. 2024). Indeed, merely speaking of the problem of misinformation makes people approach accurate information with heightened skepticism (Clayton et al. 2020; Hameleers 2023; Hoes et al. 2024; Modirrousta-Galian and Higham 2023; van der Meer et al. 2023), which may also translate to (dis)trust. As a result, audiences may become wary not only of misinformation, but also of the broader information environment, including institutions traditionally considered credible (Altay and Acerbi 2023, 2024; Altay et al. 2023a; Hameleers 2023; Hoes et al. 2024). Qualitative studies similarly suggest that the more people worry about misinformation, the less trust they report in the media and other information providers (Nielsen and Graves 2017; Van der Linden et al. 2020).

These dynamics are supported by several theoretical frameworks. First, Thorson's concept of belief echoes (Thorson 2016) explains how misinformation can shape attitudes even after people recognize it as false, suggesting that even news coverage of misinformation can have lasting, unintended effects. In her recent work, Thorson (2024) draws on a content analysis of over 800 U.S. news articles to show that misinformation coverage often emphasizes societal breakdown and platform failure,

and repeats falsehoods even when correcting them. While such coverage is typically well-intentioned, it can backfire by reinforcing public cynicism about the information ecosystem itself. Our own content analysis finds similar patterns: coverage of misinformation often features repeated falsehoods and vague attributions of blame, with corrections often buried or muted (see SI Section C.1), suggesting that media narratives about misinformation—especially when repetitive or vague—may leave audiences with lingering doubts not only about content but also about the credibility of the institutions producing or verifying it.

Second, priming theory (Van Duyn and Collier 2019) suggests that repeated cues about misinformation may activate a skeptical mindset and reshape the standards with which people evaluate information sources. Put differently, news media shape our perception of validity by making the threat of misinformation salient, which can lower trust in the very institutions that supply reliable information (Mayo 2019). In short, “Individuals who are primed to think of fake news may use fake news as a representative of news media as a whole” (Van Duyn and Collier 2019, p. 33). Mentions of false information may blur the line between unreliable and reliable content, weakening institutional confidence (Altay et al. 2024). Testing this argument, Van Duyn and Collier (2019) found that exposing subjects to tweets from journalists, political representatives, and activists about misinformation made respondents trust traditional news media less. Similarly, Tandoc Jr et al. (2021) found that telling people they have been exposed to fake news decreased trust in information institutions, including the news media. That is, these frameworks point to the risk that the discourse surrounding misinformation, however well-intentioned, may itself become a vector of epistemic uncertainty.

In sum, we argue that well-intended news coverage discussing the perils, spread, and magnitude of misinformation could make it harder for people to trust not only social and traditional media, but also the actors who produce and monitor information: scientists, academics, and fact-checkers. After all, media trust is inherently linked to trust in other institutions (Freitag and Trautmüller 2009; Lewis and Weigert 1985), especially institutions that also have an informational role (Stubenvoll et al. 2021). We thus advance the following hypotheses:

Hypothesis 3_a : *Exposure to misinformation decreases trust.*

Hypothesis 3_b : *Exposure to news coverage of misinformation decreases trust.*

Study 1: Methods

As the first step to test the hypotheses, we conducted a pre-registered,⁵ online survey experiment in the United States.⁶ We recruited respondents via Dynata using quotas based on census information on age, gender, and education level ($n = 1670$).

⁵For coherence and parsimony, in the main text, we present those pre-registered hypotheses that are also tested in Study 2. In SI Section D, we detail the results of all pre-registered hypotheses.

⁶The study received IRB approval.

Table B.3 in the SI compares our sample to the general population on a range of sociodemographics.

Stimuli Pre-test

Prior to the main study, we carried out a pre-test on MTurk ($n=205$), in which respondents rated six false statements (see SI B.3 for the statements and Table B.2 for their pre-test ratings). We note that we opted for self-fabricated claims to avoid prior exposure to these claims, which could influence the results. From all statements we picked three that were rated as similarly easy to read, interesting, easy to understand, likely to be true, and believable, and all equally plausible among Democrats ($n=119$) and Republicans ($n=68$).⁷ These statements were integrated in the stimuli and considered as ‘untrust- worthy content’ (condition 1) and ‘news coverage of misinformation’ (condition 2) in our study. We note that all selected statements contained references to the states in the U.S., which were not clearly red or blue at the time of the study, and contained no references to specific politicians or public officials. This approach minimizes the chance that our results are dependent on the specific topic, state, or politician covered in the treatments. We discuss the implications of this approach in the “[Discussion](#)” section.

Experimental Flow

After a pre-treatment survey, respondents were randomly assigned to one of two treatment groups or a control group. In the first treatment group ($n=550$), participants were exposed to *misinformation* through three professionally designed mock Facebook posts with a blurred source. We opted for a blurred source to isolate the effect of our treatment texts, and because previous research has found effects of warnings of fake news exposure regardless of the source (Tandoc Jr et al. 2021). This enhances internal validity by reducing confounds from perceived credibility or authority of the source.

These posts stated that there is a lot of information about the topic (i.e., disease outbreaks, corruption in local governments, and security breaches online), presented the false claim, and mentioned that people are finding out more about it. For example, for the topic of disease outbreak, the headline read “*There is a lot of information going around about disease outbreaks. Recently, it was revealed that local public health officials failed to report a new disease outbreak in Arizona, which may kill thousands of U.S. citizens. A growing number of people are finding out more about it.*”

In the second treatment ($n=570$), participants were exposed to *coverage of misinformation*. Three professionally designed mock Facebook posts with a blurred source stated that there is a lot of fake news about the topic (i.e., disease outbreaks, cor-

⁷ Statement 1: Local government officials in Michigan have been found to be funneling state funds to build a swimming pool on their private property; Statement 2: Local public health officials failed to report a new disease outbreak in Arizona, which threatens to take the lives of thousands of U.S. citizens; Statement 3: Nevada’s Department of Justice was recently hacked, which led to the leaking of thousands of legal documents of the state’s residents.

ruption in local governments, and security breaches online; the same topic as in the false statements in the first treatment) and mentioned that many people are trying to fight misinformation by stopping the spread of such false claims. For example, for the topic of disease outbreak, the headline read *“There is a lot of fake news about disease outbreaks. The false claim that local public health officials failed to report a new disease outbreak in Arizona, which may kill thousands of U.S. citizens, has been circulated widely. A growing number of people are trying to fight fake news by stopping the spread of such false claims.”*

This treatment operationalizes misinformation coverage as it typically appears in real-world media. Our own content analysis (see SI Section C.1) and that by Thorson (2024) shows that such coverage tends to repeat a false claim while simultaneously warning about its falsity or harmful spread. That is, these stories often emphasize the prevalence of misinformation, name a specific falsehood, and describe attempts to combat it. Our stimuli were therefore designed to reflect this structure: the posts referenced a false claim and noted that people are working to stop its spread. The control participants ($n=550$) were not exposed to any stimulus and directed to the post-test. Figure 2 presents the mock Facebook posts.

The three mock Facebook posts were presented sequentially in random order. After expo-sure, participants completed a survey assessing misperceptions and their trust in various institutions. Misperceptions were measured asking respondents to

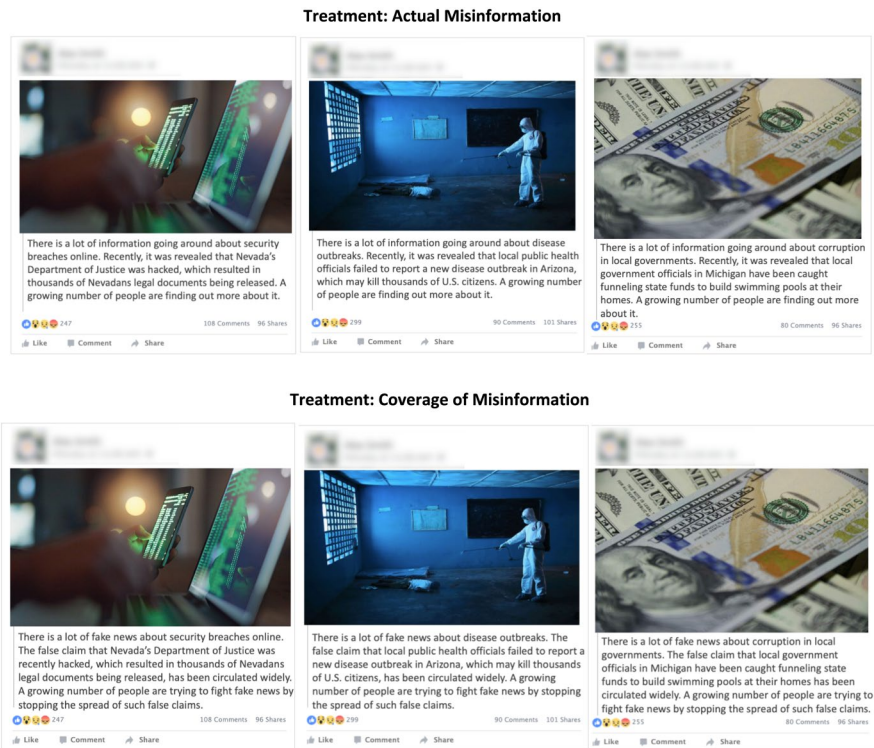


Fig. 2 Mock Facebook posts shown to participants in the experiment

indicate how accurate they think the three false statements (corresponding to the three mock Facebook posts) were on a 4-point scale (from “not at all accurate” to “very accurate”). We measured trust in seven institutions—traditional media, social media, journalists, scientists, fact-checkers and university professors—on a 7-point scale (from “I don’t trust it/them at all” to “I completely trust it/them”). We based these measures on previous research investigating the effects of political (mis)information (Altay et al. 2025a; Hoes et al. 2024). See SI B.1 for the complete pre- and post-treatment survey and the pre-registration for more precise measurement details. At the end of the survey, respondents were thoroughly debriefed (see SI B.1.1 for the debriefing).

Modeling Approach

As pre-registered, for each of the outcomes we ran a linear regression of the outcome on two treatment indicators (actual misinformation and coverage of misinformation), with the control condition as the baseline. We further adjust for a range of covariates, namely age (which of the following categories includes your current age?; 1 ‘18–24’—5 ‘65+’), gender (male, female, other), education (what is the highest level of education you have completed?; 1 ‘High School or Less’—5 ‘Masters Degree/Doctorate Degree’), political interest (generally, how interested are you in politics?; 1 ‘Not at all interested’—5 ‘Extremely interested’) and party affiliation (Please describe the option that best describes your political party affiliation; (1) A strong Democrat—(7) A strong Republican). To account for multiple comparisons, we adjust p-values (FDR) of the coefficients of interest across both studies (i.e., all coefficients for treatments in Study 1, and coefficients for exposure to misinformation coverage/untrustworthy sites in Study 2).

Study 1: Results

Figure 3 shows the coefficients for the treatment indicators in our regression model. Compared to the control, exposure to actual misinformation *increased* misperceptions by 0.30 points ($p < 0.001$, $p_{adj} < 0.001$). In contrast, participants who were exposed to the coverage of misinformation had *lower* levels of misperceptions than participants in the control and the actual misinformation group. More specifically, the coverage of misinformation reduced misperceptions by 0.24 points on a 4-point scale compared to the control ($p < 0.001$, $p_{adj} < 0.001$). The difference between seeing actual misinformation and coverage thereof is 0.54 points. Table B.4 in the SI shows all regression coefficients. These results are in line with H1 but reject H2, as only participants exposed to actual misinformation—not the coverage of misinformation—show higher levels of misperceptions.

As illustrated in Fig. 4, we found no evidence that actual misinformation or the coverage thereof has a negative effect on trust (H3_a and H3_b). These null results were remarkably similar across all information institutions examined, whether fact-checkers, journalists, or scientists, among others. Complete models can be found in SI Table B.5.1.

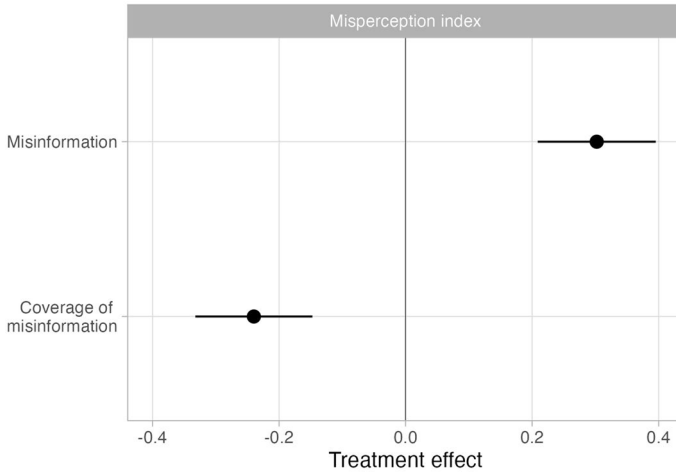


Fig. 3 Effect of treatments on misperceptions

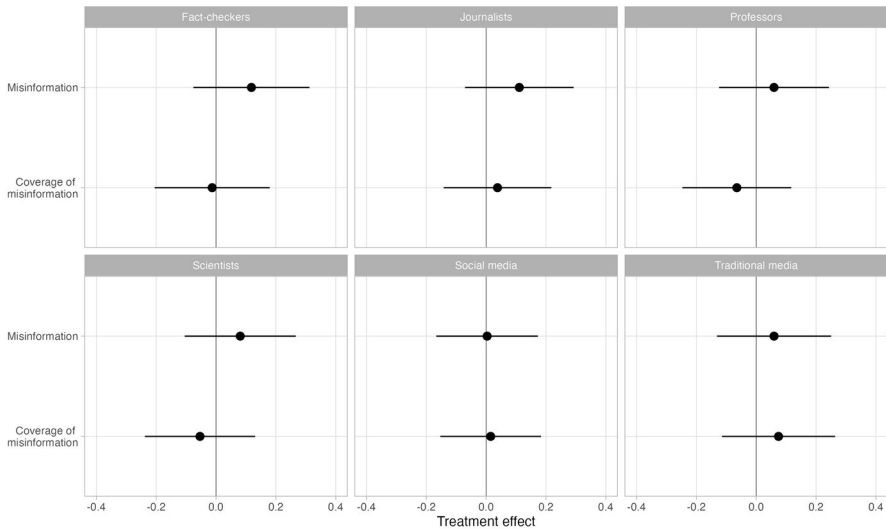


Fig. 4 Effect of treatments on trust items

In short, our experiment finds that exposure to misinformation increases misperceptions, that exposure to news coverage of misinformation lowers these misperceptions, and that neither exposure has any statistically significant effects on trust in information institutions. Although experiments are uniquely suited to establish causal effects, they are limited in terms of ecological validity, as it is not clear whether similar results would emerge in more naturalistic settings, in which users can select from nearly unlimited sources and content and where (the coverage of) misinformation is not as neatly isolated as in our experiment.

Study 2: Methods

As a second step to test the hypotheses and offer more ecologically valid evidence, we leveraged a longitudinal design that pairs individual-level self-reports with web-browsing data from two waves (W1 was conducted in August/September 2019, W2 in November/December 2019). In this study, we use respondents' web browsing data from the three months leading up to each wave. This design has advantages over both self-reported measures of (untrustworthy) media exposure as well as experimentally induced exposure, as it allows us to examine the consumption of various types of media and content in a natural setting. The panel nature of our data further allows us to control for some individual background characteristics that could confound the effect of information exposure on our outcomes, as detailed in the modeling strategy section below. However, we also point out that, as with any study using individual-level panel data, we cannot rule out unobserved time-varying confounders and therefore caution when interpreting our results causally.

To clarify the conceptual parallels between the two studies: in Study 1, we experimentally distinguish between exposure to pure misinformation and exposure to news coverage of misinformation that explicitly signals the claim is false or misleading. In Study 2, we aim to capture these two forms of exposure using behavioral data. Specifically, we use visits to untrustworthy websites as a proxy for exposure to misinformation or misleading content. We recognize that untrustworthy sites do not always publish misinformation, just as mainstream outlets may occasionally circulate falsehoods. Still, the use of domain-level proxies is common in misinformation research (Guess et al. 2020b), and we followed this convention to enable comparability with prior work. In turn, visits to articles classified as news coverage of misinformation represent exposure to journalistic content that reports on misinformation, whether mentioning fake news, specific false claims with corrective intentions, or simply discusses the prevalence or the threats of mis- or disinformation.

Participants were recruited via Lucid, an aggregator of survey respondents. We implemented quotas for age, gender, and education to match the U.S. Census margins. 1,261 adult American citizens participated in the first wave (W1), of which 948 completed the second wave (W2). Table C.13 in the SI compares our sample on a range of sociodemographics with the general population and provides tests of attrition bias between the two waves. Our analyses include those participants who completed the second wave and provided browsing data for at least seven different calendar days in the three months before the second wave ($n=804$).

Browser histories were collected using Web Historian, an open-source tool that accesses respondents' browsing histories stored on their computers, displays it to them using visualizations (e.g. network graph of websites visited, word cloud of used search terms, searchable table of browser history), and allows them to submit it to researchers (Menchen-Trevino 2016). We note that Web Historian only collects desktop browsing, and so we do not have data from mobile or tablets. We discuss this shortcoming in the discussion. The tool collects data at the visit level, which means that each visit to a web page is a record that includes a timestamp and the full URL. As some of these visits are conceptually not meaningful but nonetheless recorded as data points—for example, the browser automatically reloading a page or

people using the back button of their browser—we deduplicated visits at the person-day level. Having data at the level of the visit of a URL allows us to calculate how frequently participants visited specific news websites before completing the survey.

Classifying Browsing Behavior

Exposure to untrustworthy websites was calculated using a list of 672 untrustworthy websites created by Allcott et al. (2019), originally labeled as “fake news sites”. We use the term “untrustworthy” instead to reflect the common scholarly shift away from the politically charged and imprecise label “fake news”, and to emphasize the low credibility and misinformation-prone nature of these sites, which is the core concern in our analysis. This list combines five existing lists of untrustworthy sites that were created by previous researchers, journalists, and fact-checking sites, like Politifact. We created an individual-level count variable measuring how frequently respondents visited untrustworthy sites in the three months before each wave, which we log-transformed.

To measure *exposure to news coverage of misinformation*, we first compiled a comprehensive list of U.S. media sites based on three separate lists: the most visited 1,000 domains in the U.S. in 2018 according to Alexa, the 1,000 most tweeted domains by members of Congress in the same time period, and the 1,000 domains most often visited by our participants. We coded each domain for whether it was a news domain. We then augmented the resulting list of news domains with local news organizations as listed in public data bases. The final list containing 5,400 news domains can be found on Github (link will be made available upon publication to ensure anonymity). We then scraped the content of URLs belonging to these domains that were visited by our respondents, when possible (number of unique URLs = 146,312, after excluding 25,003 homepage, broken or defunct URLs, visits to photo galleries, user log-in pages, etc.). The text of all scraped news articles visited by our respondents was then classified.

We trained several machine learning models to predict which of the articles visited covered misinformation. We built the training set for the classifier using both random and targeted sampling (see Fig. 5). We first created an extensive list of keywords related to misinformation to do targeted sampling, rather than manually annotating a random set of articles (which would contain a small number of positive cases). Using these keywords, we selected articles that mentioned at least two of those keywords

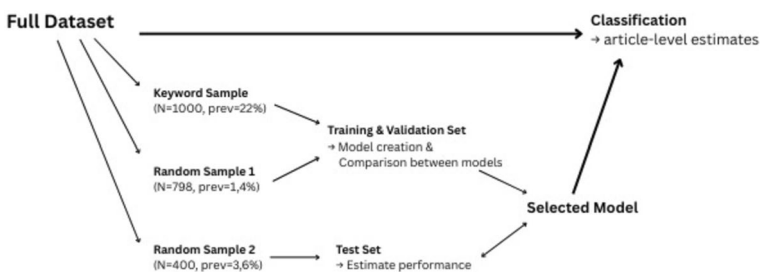


Fig. 5 Overview of the research design

from a first batch of scraped urls,⁸ to achieve a more balanced training and validation set. Then, we randomly sampled 1,000 of these articles and manually annotated them for whether they mentioned misinformation (22%) or not. Additionally, 10% were annotated by a second coder ($\kappa=0.75$). Considering the different ways, in which the media may cover misinformation, we also manually coded a subset of the articles for whether they were fact-checks, about politicians, or about content. We report these qualitative results in Section C.1 of the SI. To make sure we also include coverage of misinformation that goes beyond our keywords, we drew two additional random samples from the full corpus. We used one of these samples ($n=798$) in combination with the targeted sample for training and validating our models, whereas we reserved the second sample ($n=400$) as a test data set. In the three annotated samples, the prevalence of misinformation varies between 1.4% in the random sample and 22% in the keyword sample, given we pre-sampled for articles that contain keywords like ‘misinformation’ in the latter sample.

The binary-annotated data (covering misinformation yes/no) were used to train several classification models (see Table 1): First, as a baseline, we used the dictionary created for the targeted sample to classify whether a text covers misinformation. This provided a baseline, as one may expect articles that discuss misinformation to also include the term or its synonyms. Additionally, we trained several machine learning models that generalize from our training set to new data. We trained two bag-of-words models that rely on the aggregate frequency of words in the texts (Naive Bayes and Support Vector Machine). These models learn from the distribution of words in our training set to predict whether the articles in the validation and test sets discuss misinformation, given their distribution of words. Finally, we fine-tuned two transformer-based models, a BERT (Devlin et al. 2019) and an Electra discriminator (Clark et al. 2020) model. These models—which use self-attention to learn about language from text corpora much larger than ours and can then be fine-tuned for more specific tasks—have been successfully applied to several political science problems (Widmann and Wich 2022). Given the instability of transformers with small training sets, we initialized the model with ten different seeds. We used four epochs (i.e., iterations over the data) and kept the best interim model for each of the ten seeds based on their performance in the validation set. We present results for the best run of each model type here.⁹ The high accuracy scores (which provide the share of predictions

Table 1 Performance of dictionary and classifiers on validation and test sample

	F1 (Validation)	Accuracy (Validation)	F1 (Test)	Accuracy (Test)
BERT (best run)	0.74	0.94	0.73	0.99
Electra (best run)	0.78	0.95	0.64	0.98
Naive Bayes	0.47	0.84	0.39	0.93
SVM	0.65	0.89	0.34	0.95
Dictionary	0.45	0.60	0.30	0.87

⁸keywords: disinformation, misinformation, malinformation, misleading, fake news, fake, pants-on fire, trump-o-meter, truth, truth-o-meter, truthiness, fact-check, false, lying, snopes, politifact, factcheck.org, AFP (factcheck), post-truth, false claim.

⁹BERT can only process a maximum of 512 tokens/word embeddings per document, so we used the first 512 words of each news story for training the model as well as for generating model predictions for the

that is correct) contrasted with the comparatively lower F1 scores (which provide the weighted average of the share of correct predictions among positive predictions and the share of positive cases that is predicted) show the challenge of this task: As coverage of misinformation is relatively rare (i.e., the classes in our data are highly imbalanced), some classifiers achieve high accuracy without performing well at our task.

We evaluated the performance of the resulting classifiers on the validation set (which includes articles from the random and the targeted samples) used within the transformer model to determine model convergence, as well as our test set (which includes only articles from the random sample). These results indicate that the model meaningfully predicts the coverage of misinformation in articles. We used the model that performs best on the random sample test set (a BERT model with $F1 = 0.73$ on the test set) to generate predictions for the remaining unlabeled news articles visited by participants. We then created a count variable indicating how many articles covering misinformation a respondent visited online in the three months leading up to each survey wave, again log-transforming this count measure.

Although we did not systematically code the exact way misinformation was covered in each of these articles, our aforementioned qualitative analysis (see SI Section C.1) shows that the coverage generally aligns with what Thorson (2024) identifies as typical of U.S. news reporting on misinformation: articles mention misinformation as a broader societal problem and often cite specific falsehoods, frequently alongside corrections or fact-checks. Thus, although the format and length of real-world articles differ from our Study 1 stimuli, our measure in Study 2 captures content that reflects the same structure as our experimental treatments in Study 1 (i.e., news articles name false claims and signal that misinformation is harmful and should be addressed), increasing conceptual alignment across both studies.

Outcomes

Our two main outcomes are misperceptions and trust. We measured misperceptions by asking respondents the following question: “Do you think the following statements are false or true? Not everyone is expected to know these answers. Please just try to answer to the best of your ability. If you do not know the answer, please feel free to select “unsure” (scale from 1 (definitely false)—4 (unsure)—7 (definitely true)). Respondents were then presented with three false statements: (1) Vaccines are dangerous; (2) Having more gun owners in society reduces gun-related crimes; (3) Human activity is responsible for global warming (see C.3 for the mean belief scores for each of these statements).¹⁰ Because we are interested in whether our exposure

remaining unlabeled text.

¹⁰ We aimed to select issues that were salient at the time. In 2019, WHO declared vaccine hesitancy one of ten threats to global health and vaccine misinformation was a widely covered concern (Isaac and Frenkel 2019; World Health Organization and Commission 2019), linked to increased vaccine hesitancy, decline in vaccine uptake, and outbreaks of vaccine-preventable diseases (e.g., it was widely reported that there were around 2,000 cases of measles in 2019, compared to 372 in 2018 in the U.S. (Burki 2019)). Similarly, there were numerous mass shootings (as news media reported, “There were more mass shootings than days in 2019” (Staff 2019)), and we wanted to test misperceptions about the broader claim about guns.

variables affect general rather than specific misperceptions, we aggregated these three items into one measure. We consider participants having higher levels of misperceptions when they scored higher on the 7-point scale. Second, we measured trust by asking respondents to indicate how much they trust a number of information institutions or groups, namely scientists, university professors, and news media (operationalized through trust in reporters and journalists, scale from 1 to 7).¹¹ Because we test how exposure affects trust in specific institutions, we rely on the individual trust items. We report the full item wording and all individual trust items in Section C.3.

Modeling Approach

To take advantage of the panel structure of the data, we follow an analytical strategy discussed by Angrist and Pischke (2009) and formalized by Ding and Li (2019), who propose to model the data both with a two-way fixed effects regression (FE) and a lagged-dependent variable regression (LDV). Two-way fixed effects regression controls for time-invariant factors that vary between individuals, and time-variant factors that are constant across individuals. The main threat to causal identification are time-variant confounders that vary across individuals. In our case, this could be, for example, an increase in exposure to misinformation offline for some individuals (but not others) between the two waves. Also, it is possible that individual predispositions affect our outcomes and at the same time lead to selection into the information exposures in question may change over time and create a spurious link. In contrast, lagged-dependent variable models adjust for the outcome of the previous time point. Here, a causal effect is identified if individuals with the same value on the lagged outcome (and any other covariates adjusted for) do not differ on any other confounder. This would be the case, for example, if individuals with the same level of misperceptions in wave one have different levels of offline discussions about misinformation before wave two. Ding and Li (2019) show that the two models can “bracket” the true effect: If the identification assumption of LDV is correct, FE will overestimate the effect; if the identification assumption of FE is correct, LDV will underestimate the effect. Therefore, we present the results from both models side-by-side, adhering to best practices and increasing result transparency. We acknowledge that the causal identification assumptions for both models may be overly optimistic, and therefore refrain from interpreting our results strictly causally.

A benefit of our data is that—unlike in most panel situations—the independent variables (i.e., behaviorally tracked exposure) are measured *before* the outcomes (i.e., survey measures). Thus, we can rule out that the outcome affects the predictor *in the same wave*. However, it is plausible that web browsing is influenced by outcomes of *a previous time period*. Such bi-directional influences over time can be accounted for by cross-lagged panel models. However, this approach has come under criticism for

Lastly, in 2019, Donald Trump questioned climate change repeatedly, including his famous tweets that questioned climate change due to unusually cold weather in the U.S. and that claimed that global warming was a hoax perpetrated by China (Worland 2019). We address a few considerations related to the items in the discussion.

¹¹ We note that this study is part of an original, larger project which measured trust in several additional institutions, from which we select informational institutions only.

not separating within- and between-subjects variation (Hamaker et al. 2015). More sophisticated approaches unfortunately require more than two waves of data.

With these caveats in mind, we control for potential confounders. First, we control for a couple of time-variant factors. Because differences in the exposures to both untrustworthy websites and news coverage of misinformation could reflect differences in general browsing frequency (in that those who are more active online have more chances to encounter both untrustworthy sites and news coverage of misinformation), we control for a person's overall visits in the period (log-transformed). Next, as many news organizations in the U.S. are partisan and the ideological leaning of the U.S. media has been found to correlate with misinformation frequency (Benkler et al. 2018; Grinberg et al. 2019), we also control for the partisan nature of individual news exposure. Toward this end, we rely on the domain-level scores developed by Robertson et al. (2018). These scores are based on the Twitter linking patterns of partisans and were cross-validated with self-reported data and other methods of measuring a domain's political leaning. A lower score indicates that the domain has a more liberal audience, and higher scores indicate more conservative audience. We transformed the continuous scores into a categorical variable indicating liberal ($n=345$) and conservative ($n=470$) domains, using cutoffs adapted from previous work (Weeks et al. 2021), and included them as covariates in our analyses. All the news domains on our list had a corresponding score by Robertson et al. (2018). We refer the reader to Github (to be made available upon publication) for the domain ideology scores.

The LDV models also require controlling for time-invariant factors. We include age, gender and educational attainment (coded as a dummy for high education) as covariates. As political misperceptions and trust are driven in large part by partisan biases (Hooghe 2020; Nyhan 2020) and as our survey misperceptions items may represent partisan identification or engrained political attitudes, we also include one's party affiliation (a 7-point scale from "A strong Democrat" to "A strong Republican"). Finally, we control for political interest ("How interested would you say you are in politics?", 7-point scale) and self-reported knowledge, measured with a five-item battery (e.g., "Among my circle of friends, I'm one of the 'experts' on politics.", 7-point scale) which we averaged into an index (" $\alpha=0.88$ "). Finally, as in Study 1, we account for multiple comparisons by adjusting p-values (FDR) of the coefficients of interest across both studies.

Study 2: Results

We first turn to our observational data to describe the prevalence of exposure to untrustworthy sites and news coverage of misinformation. In line with previous work, we find low levels of exposure to untrustworthy sites (which are not classified as news domains), namely 6,321 visits across the two waves (3,448 in the first wave and 2,873 in the second wave), with an individual-level average of just 7.9 visits (4.3 in the first and 3.6 in the second wave). Again, this is a highly skewed distribution (see SI Figure C.1). In addition, of the 638,434 visits to news domains across the two waves, only a fraction—a mere 0.4 percent—covered misinformation (2,789 across the two waves, 1,433 in W1, 1,356 in W2). On average, an individual

had 3.5 visits to misinformation coverage across the two waves (1.8 in the first wave and 1.7 in the second wave). As shown in Figure C.1 in the SI, this distribution is highly skewed, with few individuals accounting for most of this type of exposure. Overall, although news media reporting of misinformation has increased (as we show in Fig. 1), this does not seem to translate into frequent exposure to such reporting at the individual level, at least not in the present sample. These descriptives suggest that experimental designs, which rely on “forced exposure” to certain content—whether misinformation or news coverage of (fact checked) misinformation—are far removed from naturalistic settings, in which most citizens do not encounter (the coverage of) misinformation in such an isolated manner.

We use the behavioral data to predict respondents’ levels of misperceptions and trust (see SI C.3 for descriptive statistics). First, we use an indexed measure of misperceptions (i.e., belief in the three false statements) as an effect of participants’ exposure to untrustworthy sites as well as the coverage of misinformation, modeled with both the lagged-dependent variable (LDV) and fixed-effects (FE) approaches described earlier. Positive coefficients indicate the predictor increases misperceptions. In SI C.5, we present the regression tables with the results, including the effects on the individual misperceptions items.

Figure 6 plots the coefficients of predictors of participants’ misperceptions. The exposure variables, neither untrustworthy sites nor news coverage of misinformation, do not predict increased misperceptions, which is contrary to H1 and H2. There are also no statistically significant effects of these two types of coverage on the individual misperception items (see Tables C.15 and C.16).

Turning to the effects on trust, Fig. 7 plots the coefficients predicting individual trust from exposure to untrustworthy sites and news coverage of misinformation (see Tables C.17 and C.18 for full results). First, we find suggestive evidence that exposure to untrustworthy sites decreases trust in some information institutions (i.e., news media, scientists). These patterns, however, mostly emerge in fixed effects models, not in LDV, and most coefficients become statistically insignificant at the 95%-level once we apply FDR adjustment. To exemplify the more consistent effects

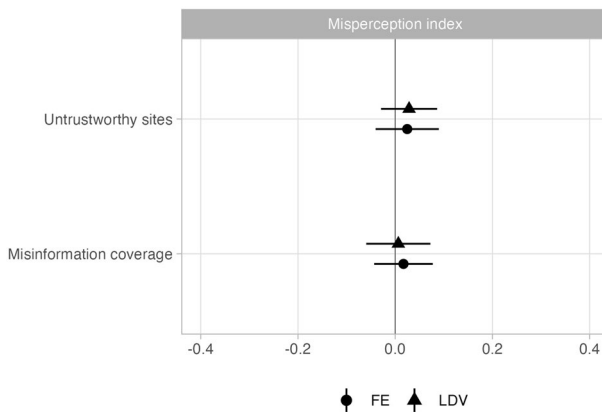


Fig. 6 Predictors of misperceptions

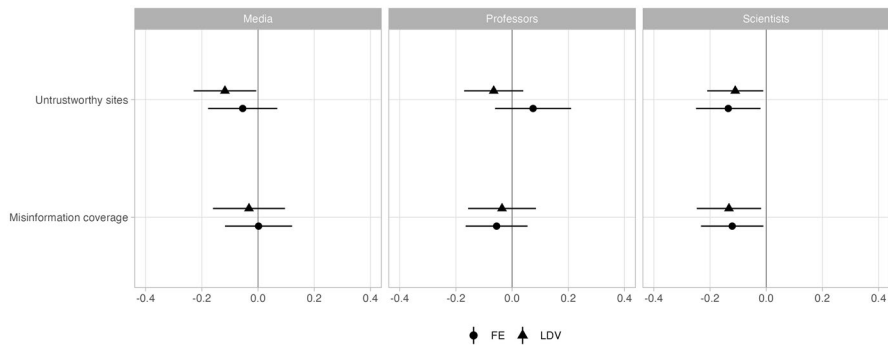


Fig. 7 Predictors of trust

across both models, visits to untrustworthy sites predict decreased trust in media (LDV: 0.12 points on a 7-point scale, $p < 0.05$, $p_{adj} = 0.25$; FE: 0.05 points, $p = 0.38$, $p_{adj} = 0.82$) and scientists (LDV: 0.11 points on a 7-point scale, $p < 0.05$, $p_{adj} = 0.24$; FE: 0.14 points, $p < 0.05$, $p_{adj} = 0.24$). In sum, we find partial evidence in favor of *H3a*.

When it comes to exposure to news coverage of misinformation, we find some evidence that encountering news coverage of misinformation predicts decreased trust in scientists. Similarly to untrustworthy sites, the effects of the coverage of misinformation become statistically insignificant after the FDR adjustment. Still, the decrease of trust in scientists remains statistically significant across both models (LDV: 0.13-point decrease on a 7-point scale, $p < 0.05$, $p_{adj} = 0.24$; FE: 0.12-point decrease on a 7-point scale, $p < 0.05$, $p_{adj} = 0.24$). Coefficients for media and professors are smaller and less consistent across model specifications. We conclude that there is limited evidence that news coverage of misinformation decreases trust in information institutions overall, with the exception of a robust decrease in trust in scientists (*H3b*).

Taken together, we see a suggestive pattern that untrustworthy sites and news coverage of misinformation have no effects on misperceptions but decrease trust in some information institutions. That said, these results are not very robust apart from the finding that both untrustworthy sites *and* the coverage of misinformation decrease trust in scientists, where both the LDV and FE approaches yield significant coefficients.

When it comes to the covariates, in the models presented in the main paper as well as in those testing the individual misperceptions items (see Section C.5 in the SI), partisanship strongly predicts politically congruent misperceptions; self-reported Republicans were more likely than Democrats to believe that vaccines are dangerous, that having more gun-owners in society reduces gun-related crimes, and that human activity is not responsible for global warming. We observe a similar pattern for trust, with the more Republican participants having lower trust in the media, scientists, and university professors than the Democrats. These findings confirm that partisanship is an important factor predicting misperceptions and trust. The coefficients for the other socio-demographic covariates were not statistically significant.

Conclusion and Discussion

In this two-study project using both experimental and online behavioral data, we test the effects of untrustworthy sites and news coverage of misinformation on citizens' mis-perceptions and trust in several key information institutions. We advance the argument that media attention to misinformation may have unintended consequences, and that, next to misperceptions, one should consider trust in media and scientific actors as a crucial outcome when studying the effects of these different types of exposure. Overall, the results from Study 1 and 2 are not always consistent, underscoring—on the one hand—the challenges in capturing the effect of misinformation and news coverage thereof and also—on the other hand—the fact that the detected results are highly dependent on the methodologies used.

First, the experimental Study 1 supports the concerns that exposure to untrustworthy content may exacerbate misperceptions among the American public (Guess et al. 2020b; Murray et al. 2020; Pennycook and Rand 2019; Silverman and Singer-Vine 2016). That said, our behavioral trace data from Study 2 do not reveal any effects of actual exposure to untrustworthy sites on misperceptions, which also aligns with evidence for null- or limited effects of misinformation or untrustworthy sites (Altay et al. 2023b; Valenzuela et al. 2022; Weeks et al. 2021). We speculate that the mixed findings, which add to the already mixed evidence from extant research, are due to differing methodologies.

Study 1 showed three false statements and asked participants about these statements immediately after exposure. Such short exposure increased belief in self-fabricated false claims that were ideologically neutral and about 'new' issues, events, or actors. Naturally, such designs are more likely to find significant effects, especially when individuals have no reason to disbelieve the claims. We intentionally opted for self-fabricated false claims and blurred sources to minimize the likelihood of prior exposure or partisan cuing, which could confound effects. This approach allowed us to isolate the role of content familiarity and misinformation exposure, independent of political identities and sources. However, we recognize this design choice may limit ecological validity in that misinformation often circulates in familiar formats from familiar sources and draws on recognizable events. Our fabricated and "source-less" false claims lack such contextual embedding, which strengthens internal validity but may not fully capture the complexity of real-world misinformation exposure.

Study 2, in turn, captured actual visits to websites categorized as untrustworthy, finding null effects on individual misperceptions about actual political issues assessed up to six months after exposure. The null effects may be due to the fact that the untrustworthy sites were a small drop in any individual's online diets, the time that had elapsed between exposure and survey measurement, the fact that the survey assessed misperceptions about salient issues related to partisanship, political knowledge, or political interest (we note, however, that the null effects emerged when controlling for these factors), and/or because the articles the participants visited may have been about other issues and not about vaccines, gun control, or climate change (i.e., the three misinformation claims assessed in the surveys), as we note below.

For now, we speculate that our evidence from online trace data more accurately portrays the limited effects of untrustworthy websites as occurring "in the real world",

given that infrequent visit to an untrustworthy site may be overshadowed by numerous other visits to news and entertainment websites, generating limited over-time effects. That said, it is possible that untrustworthy websites *do* influence misperceptions, but primarily among their most frequent users, who were likely underrepresented in our sample. Inasmuch as individuals who visit these sites are likely to endorse conspiracies and be very conservative (Calvillo et al. 2020), they are unlikely to partake in academic studies, especially those that require them to share their browsing history. We encourage researchers to oversample the users of interest (Chen et al. 2023). Such additional efforts are needed to disentangle the likely mutually reinforcing relationship between misperceptions and self-selection to misinformation and untrustworthy websites.

Our second key finding from the observational data is that untrustworthy sites lower trust in some information institutions due to the skeptical and sometimes mocking coverage (e.g., questioning experts, referring to legacy media as ‘fake news,’ etc. (Egelhofer and Lecheler 2019; Eldridge II 2019; Elswah and Howard 2020; Paul and Matthews 2016; Rae 2021)). In our case, this decreased trust emerged particularly for scientists with more limited effects for reporters and journalists as well as university professors. We expand on this below. In contrast, experimentally manipulated exposure to untrustworthy content did not decrease trust. This is likely because trust is a more stable factor, which builds over time and is less easily affected by one-shot exposure to three pieces of content than is accuracy with which people evaluate information in these specific pieces of content.

Third, Study 2 shows that news coverage of misinformation—which in our observational data includes fact-checking efforts on sites such as Snopes and Politifact—has unintended spillover-effects that may add to those of direct exposure to untrustworthy sites. We argued that the more misinformation is discussed and the more news media emphasize its perils, spread, and magnitude, the harder it may become for people to know whom to trust (Altay and Acerbi 2024). We find that news coverage of misinformation lowers trust in the very “source” that should provide evidence to counter false claims: scientists, with less consistent effects for the two other information institutions tested (professors and news media). This finding aligns with past work (Tandoc Jr et al. 2021; Van Duyn and Collier, 2019) and reinforces general concerns about both misinformation and also news coverage thereof creating epistemic uncertainty (Thorson 2024).

Although we did not pre-register hypotheses specific to scientists, this group is especially likely to suffer trust erosion following exposure to misinformation and its coverage. Scientists have been at the center of many misinformation debates—ranging from vaccines and climate change to COVID-19—making them a frequent subject of both untrustworthy content and journalistic discussions about misinformation. Moreover, scientific communication involves uncertainty, evolving knowledge, and probabilistic claims, which can appear ambiguous or unconvincing to the public. In the context of news attention to misinformation, these characteristics may inadvertently signal unreliability rather than expertise, especially when framed through narratives that emphasize doubt or conflicting information. These dynamics may help explain why scientists emerge as particularly vulnerable to trust erosion. We encourage future work to look more in-depth at the content of untrustworthy

sites and at news coverage of misinformation to uncover what specific institutions or political groups are targeted. Given the declining trust in science in the contemporary U.S., this finding and its extensions are timely.

Together, these effects of untrustworthy content and of media coverage of misinformation underscore the importance of employing different methods and the difficulty of interpreting the contradicting results they yield. Experimentally, single exposure to (news discussing) ideologically neutral false claims one has never encountered before may be sufficient to trigger belief in these claims, but not enough to shift trust. In turn, in naturalistic settings, visits to untrustworthy sites as well as reading news about misinformation, fact-checking, and terms “fake news” may not influence misperceptions on specific topics assessed but could reduce people’s ability to discern between what is true and false in general.

Last but not least, our descriptive analysis from the six months of online behavioral data echoes past findings that exposure to untrustworthy sites is limited (Aslett et al. 2022; Guess et al. 2019): on average, people in our sample visited untrustworthy sites about 3.5 times in the three months leading up to wave 2. Exposure to the coverage of misinformation seems to be rarer in our sample, despite the fact that there have been remarkable increases in the use of the terms “fake news”, “misinformation,” and “disinformation” in major U.S. newspapers over time. On average, people in our sample read news content covering misinformation only 1.69 times over six months. This is an important descriptive finding for future research on the potential spillover effects of the coverage of misinformation (e.g., (Jungheer and Rauchfleisch 2022): actual exposure to such coverage—despite its aggregate increases—is not as common as assumed. Thus, it is yet more noteworthy that the effects of exposure to the coverage of misinformation occur above and beyond those of visits to untrustworthy sites, as the two exposures do not overlap in our data.

We acknowledge that we may be underestimating exposure to misinformation or to the news coverage thereof. For one, our observational data only account for a fraction of people’s overall news diet, as we cannot observe news consumption offline or on mobile devices. Television is still the most dominant news source for most people (Allen et al. 2020), and it may well be that our participants were exposed to (news coverage of) misinformation in these other information environments. Second, our data cannot capture inadvertent exposure but only active clicks on articles or content. Furthermore, because we only record visits to untrustworthy sites from a pre-defined list, we omit the—potentially much larger—exposure to misinformative posts, memes, or videos produced by individual users, social media groups, or other accounts that are not on the original list. In short, we cannot capture offline communication and passive exposure to misinformation or news coverage thereof, which may happen while watching television or YouTube, scrolling through a social media feed, visiting a news homepage, or even chatting online with family members on WhatsApp. Capturing all these exposures and all the different—and fast evolving—forms of misinformation (including deep fakes or AI-generated content) presents an ever-pressing challenge to researchers.

Despite the significance of these results, our studies are not free from limitations. As aforementioned, in Study 1, we intentionally opted for self-fabricated false claims and blurred sources to minimize the likelihood of prior exposure or partisan cuing,

which could confound effects. We recognize that this design choice may limit ecological validity and does not fully capture the complexity of real-world misinformation exposure. Similarly, we opted for treatments that contained no references to specific politicians. Yet, news coverage of misinformation—including fact-checks—may harm trust in actors most when they are actively blamed (Hoes et al. 2024). Although meta-analyses of fact-checking found that their effectiveness does not depend on whether the debunked falsehood targets specific actors versus general false claims (Chan et al. 2017; Walter et al. 2020), specific blame attributions may matter more when it comes to trust. We encourage work that systematically varies the extent to which (the coverage of) misinformation targets specific (political) actors. In short, our focus was not on the interaction between content and source, but on contrasting misinformation versus its coverage. Many studies already examine the effects of source trustworthiness; our aim was to contribute to a growing body of work that isolates how different content types—independent of source—affect public beliefs and trust.

We also acknowledge a few limitations of Study 2. Although we used visits to untrustworthy sites as a proxy for exposure to misinformation, we did not examine whether each of the 6,321 untrustworthy websites visited by the participants covered misinformation in general and discussed vaccines, gun control, or climate change in particular (topics of our misinformation claims in the surveys). Because these three topics were prominent in the news during data collection, we suspect that they were covered by the untrustworthy websites, yet we cannot ascertain how often. Similarly, the measurement of the *coverage of misinformation* at the article level introduces some insecurity. By parsing articles from a large number of domains, we necessarily trade some degree of quality of the extracted texts for the size of the corpus. The automated classification of articles that discuss misinformation posed considerable challenges, given the low prevalence of these articles. While the performance of our classifier is significantly better than standard dictionary and bag-of-words approaches, it may still miss some articles that discuss misinformation (sensitivity), while mis-classifying articles that do not do so (precision).

In addition, our misperception measures in Study 2 could reflect individual partisanship or ideology. We account for this by controlling for party affiliation in our LDV models. We also note that these misperception items primarily capture right-leaning misinformation. This is in part because there is more right-leaning misinformation on the supply side (Mosleh and Rand 2022) and the political right is more likely to produce, endorse, and share misinformation (González-Bailón et al. 2023; Grinberg et al. 2019; Mosleh et al. 2024), decreasing our ability to capture misinformation produced or endorsed by the political left. In short, we acknowledge that our misperception items are not representative of the range of false claims in the information environment and a better approach would have been to select specific false claims circulating at the time of the study. We encourage future research to replicate our findings using different topics and issues.

Lastly, we note that very few people seek out news or go online for politics (Wojcieszak et al. 2024) and so—by capturing active consumption of untrustworthy websites and/or news articles covering misinformation—we mostly speak to exposures among those internet users who self-select to these two kinds of news websites.

This group is more vocal and participatory than the general population, which—as recent research suggests—tends to be politically disengaged (Krupnikov and Ryan 2022) and averse to politics (Klar et al. 2018).

Despite these limitations, which offer a fruitful area for future research, our study is the first to explore the effects of exposure to untrustworthy sites and news coverage of misinformation side by side and across two studies, testing their distinct effects on misperceptions and trust with two different types of data. We conclude that the effects of misinformation and its coverage are highly dependent on the type of data used and the specific design of experimental treatments. Moreover, these effects are bound to be context-specific and influenced by media and party systems (among many other factors). As our findings only speak to the U.S., a highly polarized two-party system with a unique media environment, future work should extend this research to other countries to shed more light on the potential effects of misinformation, untrustworthy sites, and news coverage of misinformation on the decline in trust. Systematic multi-method and multi-design comparative work on these effects is needed to better understand the role and the implications of misinformation, untrustworthy sources, and legacy media covering misinformation, fake news, and fact-checking. Given the challenges faced by many democracies around the globe, such research is both timely and needed, and it is our hope that the evidence provided here opens an important methodological and substantive discussion on these issues.

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Data Availability The data, code, and materials required to replicate the analyses reported in this article, as well as the study pre-registration, are publicly available via the Open Science Framework at <https://osf.io/xq7wp/files/osfstorage>.

Declarations

Competing interests M.W. is part of the US 2020 Facebook and Instagram Election Study, which is a pro bono and fully independent research collaboration of researchers with Meta. The remaining authors declare no competing interests.

Ethical approval This study was approved by the Institutional Review Board at the University of California, Davis. Informed consent was obtained from all participants prior to data collection. The study was conducted in accordance with the ethical standards outlined in the *American Political Science Association’s Principles and Guidance for Human Subjects Research*.

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